

Workflow Notes and Constraints - Mercer General ED Triage Process

Patients arrive at the ED and are screened for life-threatening conditions. If a patient shows a CTAS 1, they bypass the registration and go directly to resuscitation. All other patients proceed to register, noting that in the Caribbean ED context, stated by (1), registration and triage areas are separate, which introduces an additional delay.

Following the registration, the triage nurses conduct a triage assessment, this is done simultaneously by taking a history and capturing vital signs. This is consistent with (1) observation that triage was integrated with patient streaming. The nurse then assigns a CTAS level (1-5), streaming patients to either the critical area or wait/minor area.

After streaming, the ED clinicians then conduct a full assessment before doing their diagnostics. Once a decision is made, patients are either discharged, admitted to the ED ward or referred to the inpatient specialist team to be reviewed before transfer or exit.

3 Constraints

1. Paper-Based Vitals

Patient vitals are recorded on paper before an electronic health record (EHR) entry, creating documentation delays. This limits the real-time data availability that any machine learning (ML) triage tool depends on.

2. Nursing Shortages on Night Shifts

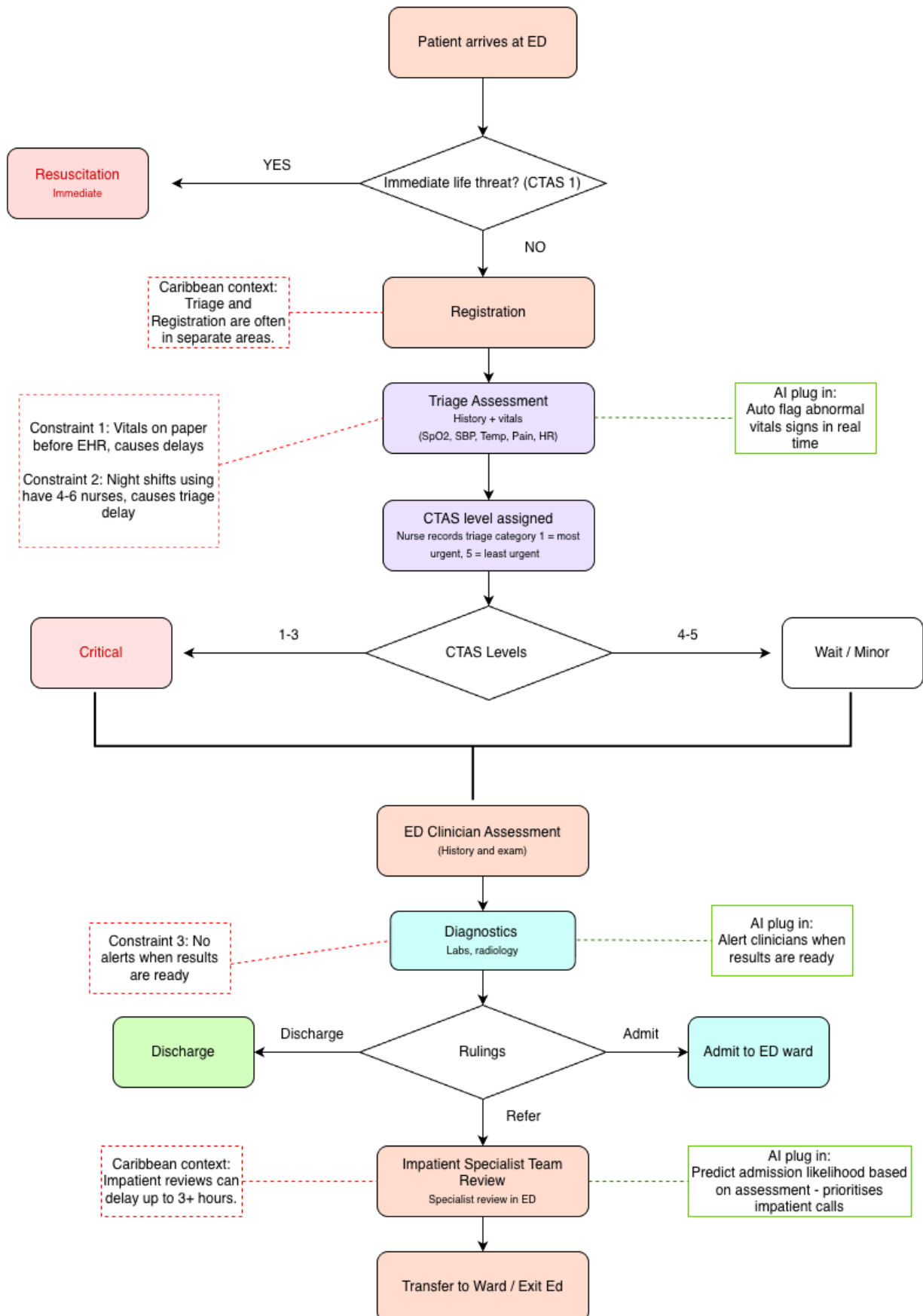
(1) found about 4-6 nurses on shifts that required 14. Any AI tools that add additional steps or data entry at the triage desk will be abandoned under those conditions.

3. No Alert Systems for Diagnostic Results

Without any notifications, clinicians physically collect reports from other departments. (1) identified this as a direct contributor to patient flow delays.

The flowchart below illustrates the full triage process at Mercer General ED, annotated with AI plug-in points and Caribbean specific workflow constraints.

Mercer General ED - Triage Process Flowchart



References

1. De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open*. 2020 Dec 12;10(12):e041422. doi:10.1136/bmjopen-2020-041422 PubMed PMID: 33310804; PubMed Central PMCID: PMC7735135.